

Demystifying Research Methods course manual

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1. Introduction

Demystifying Research Methods provides you with an essential toolkit for conducting applied research in the field of Safety and Security. You will develop a comprehensive understanding of the entire research process, from initial analysis to final design. By the end of the course, you will be able to:

- Apply core research concepts and variables to real-world security situations.
- Conduct preliminary research on a topic using frameworks like the 6W problem analysis.
- Critically evaluate the quality of research questions, research designs, and potential limitations in existing studies.
- Formulate your own applied research design.
- ... and more. See all the [Learning Goals](#) for this course in the next section.

The skills you build here are fundamental to your success. They will prepare you for writing assignments, reports, your final bachelor's thesis, and for solving practical problems throughout your career. As the first course in the Research Methods track, it lays a strong foundation for the rest of your studies.

2. Learning Goals

The learning goals for this course are connected to specific assignments and to the overarching SSMS competencies, which you can find in the SSMS programme prospectus.

- 1) The student **applies** basic applied research concepts to different safety and security situations and contexts.
- 2) The student **conducts** basic preliminary research on a safety and security topic using the 6W problem analysis framework on various types of literature.
- 3) The student **distinguishes** the main concepts and variables of a safety and security study, as well as examines their connections.
- 4) The student **evaluates** different research questions, in terms of their type, clarity, focus, relevancy, feasibility, complexity, bias, and ethicality, within different safety and security contexts.
- 5) The student **appraises** the research design, including the research approach, research problem, objective, central question, and sub-questions of different safety and security studies.
- 6) The student **appraises** safety and security studies in terms of potential limitations, such as data availability or access, sample size or representativeness, generalisability of results, time or budget constraints, and other relevant limitations.
- 7) The student **formulates** their own applied research design, including a research problem, objective, central question, and two (2) sub-questions for a safety and security topic.

Learning Goals	Linked to SSMS competencies
1) The student applies basic applied research...	2, 4, 5, 7
2) The student conducts basic preliminary research...	1, 2, 4, 5, 7
3) The student distinguishes the main concepts...	2, 4, 5
4) ..., 5) ..., 6) ..., 7) ...	1, 2, 4, 5, 7

3. Course literature and materials

All required materials for this course are available on [Brightspace](#). The weekly programme will guide you through the specific resources you need for each session, which include:

- PowerPoint slides;
- Workshop and homework exercises;
- Concept lists with explanations, extra materials, and videos;
- All the DRM quizzes;
- Selected excerpts from research methods textbooks.

The primary textbook for this course is “Doing Research: The Hows and Whys of Applied Research by Verhoeven”. Both the 5th and 6th editions are acceptable. You can access the book in several ways:

- Online: Available for free through the THUAS/HHS Library portal;
- On-site: Limited physical copies are available at the THUAS/HHS Library;
- Purchase: Second-hand copies can often be purchased from SSMS graduates.

Note on terminology: Research methods are used for different purposes, by researchers and practitioners in many different fields, so you may encounter slight variations in how concepts are defined. For all your assessments in this course, prioritise the information and definitions provided in the required course materials on Brightspace.

4. Exam structure and content

This course is assessed with a cumulative exam delivered in three parts throughout the semester, and three quizzes assessed as Pass or Fail. Your final grade is based on your total performance across all three parts of the cumulative exam and whether you passed the three quizzes.

Component	Details
Exam type	A cumulative exam with multiple-choice (MCQ) and open-ended questions, divided into three parts (CT1, CT2, CT3). Three quizzes (Q1, Q2, Q3).
Exam materials	All required literature and materials provided on Brightspace up until the date of the exam, including slides, exercises, quizzes, and readings.
Format	Consult the appendices for more information about the format of each exam part: Part 1: CT1 , Part 2: CT2 , and Part 3: CT3 . Take note also of the resit of the three parts , as well as the criteria used for the open-ended questions and the quizzes .
Cut-off mark	The cumulative exam must be completed successfully, with a grade of 5.5 or higher (1-10 scale) to pass the course. The number of points needed for the 5.5 is variable and adjusted for difficulty . The three quizzes must receive a Pass.
Final grade weight	Part 1: CT1: 20%, Part 2: CT2: 50%, Part 3: CT3: 30%, Resit: 100%. Quizzes: Pass/Fail
Assessment dates	Consult your official timetable for the most up-to-date schedule.

5. Grading and resit information

Please note that, the points awarded for each exam part are **scaled to a course total of 1000 points in the Osiris system** (CT1 = 200, CT2 = 500, CT3 = 300). Your final grade is calculated based on this total. **A strong performance in one part can compensate for a weaker performance** in another (e.g. CT1 is only worth 20%, so it can easily be compensated with a high score for CT2 and CT3).

Because this cumulative exam contains MCQ, **a guessing factor is automatically applied**, which takes into account the chance of correctly guessing an answer. This guessing factor is automatically calculated (and applied) by Remindo, based on the number of options and combinations available. If a score is **below this guessing factor, the grade is automatically a 1/10**.

The number of points you need to pass, **the cut-off score, is corrected for difficulty** based on the overall performance of all students. Instead of the traditional 55% of the max. possible score (i.e. 5.5/10), the cut-off score is based on 60% of the top 5% highest scores for the exam; that number becomes the new 5.5/10 for that particular exam, and the average of those 5% highest scores becomes the new 10/10. Because of this, the cut-off score for this exam can only be calculated at the end of the three testing moments, when all students have their total score. **Once all testing moments are complete, the sum of points obtained will be converted into a grade of 1-10**.

In accordance with THUAS *cumulative* assessment procedures, it is not possible to resit individual parts of an exam (i.e. individual parts such as the CT1, CT2, or CT3). **The resit is a single, comprehensive exam (120 minutes) covering all course material (CT1, CT2, and CT3 in one exam)** and counts for 100% of your final exam grade. **The cut-off score for the cumulative exam resit is the same as for the first opportunity of the same exam** (i.e. the cut-off score that was calculated based on CT1, CT2, and CT3), which is explained above. The points score of the resit is converted into a grade of 1-10 shortly after the exam is taken.

The three quizzes are assessed via Brightspace, using a Pass or Fail rubric. These quizzes prepare you directly for the open questions in the cumulative exam and are meant to help you track your progress throughout the semester. **You must receive a Pass for the three quizzes, or their resits**. You are encouraged to use any remaining quizzes as preparation for the exams.

6. Weekly programme

This schedule provides an overview of the topics and activities for the semester. Note that **you must only attend your assigned workshop slot each week**, as work will be conducted in assigned groups. Bring writing materials, or a laptop/ mobile device. All **preparation materials are on Brightspace**.

Session	Content and key activities	Preparation
Week 2 – Lecture 1: Intro to Research Methods (part 1)	• Role of applied sciences in safety and security.	Read course manual.

Week 3 – Workshop 1: Intro to Research Methods (part 2)	Group work: applying core concepts to short SSMS cases.	Study materials on Brightspace (prioritise concept list).
Week 4 – Quiz 1: Intro to Research Methods (part 3)	- Quiz about core concepts. - Preparation for CT1.	Study all materials so far, including all homework.
Week 5 – Lecture 2: Understanding the problem (part 1)	6W problem analysis framework and sources. - Concepts, variables, and relationships.	Study all materials so far.
Week 6 – Workshop 2: Understanding the problem (part 2)	Group work: applying the 6W framework.	Review previous sessions.
Week 7 – Quiz 2: Understanding the problem (part 3)	- Quiz about 6W problem analysis framework. - Preparation for CT2.	Study all materials so far, including all homework.
Autumn Break		
Week 8 – Quiz 1 resit: Intro to Research Methods (part 3)	- Quiz about core concepts. - Preparation for CT1.	Review previous sessions. Study all materials so far. Complete quiz 1.
Week 9 – Quiz 2 resit: Understanding the problem (part 3)	- Quiz about 6W problem analysis framework. - Preparation for CT2.	Review previous sessions. Study all materials so far. Complete all quizzes and resits so far.
Week 10 – CT1 test (20%)	All content from previous sessions.	
Week 12 – Lecture 3: Planning your investigation (part 1)	Building an applied research design (ARD) and identifying limitations.	Review previous sessions. Complete all quizzes and resits so far.
Week 13 – Workshop 3: Planning your investigation (part 2)	Group work: create and assess an ARD.	Review previous sessions.
Week 14 – Quiz 3: Planning your investigation (part 3)	- Quiz on building an ARD and identifying limitations. - Preparation for CT2.	Review previous sessions. Study all materials so far, including all homework.
Week 15 – CT2 test (50%)	All content from previous sessions, focus on 6W and ARD.	
Week 17 – Lecture 4: Integrating the whole process	- Evaluating research choices across a full case. - Advice for Research Problem next semester.	All content from previous sessions.
Week 18 – Quiz 3 resit: Planning your investigation (part 3)	- Quiz on building an ARD and identifying limitations. - Preparation for CT2 resit.	Review previous sessions. Study all materials so far. Complete all quizzes and resits so far.
Week 20 – CT3 test (30%)	All content from previous sessions.	
Semester 2, Week 5/6 – Exam resit (100%)	All content from previous sessions.	

Post-course: Review session for all semester 1 exams	• Opportunity to review your exam performance. Details will be announced on the student portal on Brightspace.	
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7. Rules and regulations

Attendance at lectures and workshops is not mandatory but highly recommended. The content covered in these sessions directly prepares you for the exams. All classes are conducted in-person on campus. Unless announced otherwise, livestreams or recordings will not be provided.

You are expected to refer to Brightspace regularly for updates about the course content, schedule, assessment details, or other announcements. It is your responsibility to stay informed.

All individual assignments and tasks must be completed without the assistance of unauthorised tools or other people. Using AI tools (e.g. ChatGPT, Gemini) for tasks that are meant to be completed individually is considered a breach of academic integrity. Keep in mind that you will not have access to these tools during exams, and that their output often fails to meet the required academic standards.

At THUAS, additional time, resources, and guidance are made available to students with disabilities. Upon consultation with your assigned Academic Advisor, accommodations can be made as needed to ensure that you can fully participate in the course and its assessments. Accommodations cannot be made unless this official process has been followed.

APPENDIX 1: Exam part 1 (CT1) – Assessment details and matrix

Component	Details
Exam part	CT1: Cumulative testing moment 1
Exam materials	All required literature and materials provided on Brightspace up until the date of the exam, including slides, exercises, and readings.
Format	20 multiple-choice questions.
Duration	30 minutes (conducted via Remindo).
Final grade weight	20% of the final course grade (max. 200 points).
Cut-off grade	Calculated after all three testing moments have been completed, and the exam is adjusted for difficulty.
Assessment dates	Consult your official timetable for the most up-to-date schedule. 1 st opportunity: Semester 1, November. 2 nd opportunity: Semester 2, March-April (as part of the full resit).

The primary emphasis of this exam is on **Applying** concepts (75% of questions), with the remainder focused on **Analysing** (20%) and **Evaluating** (i.e. appraising, 5%) research components.

Learning Goals assessed	Questions	CT1 weighting
1. The student applies basic applied research concepts to different safety and security situations and contexts.	15 MCQ (1 pt each)	15 pts = 75%
4. The student evaluates different research questions, in terms of their type, clarity, focus, relevancy, feasibility, complexity, bias, and ethicality, within different safety and security contexts.	1 MCQ (1 pt each)	1 pt = 5%
5. The student appraises the research design, including the research approach, research problem, objective, central question, and sub-questions of different safety and security studies.	4 MCQ (1 pt each)	4 pts = 20%
Total:	20 questions (20 pts)	100%

APPENDIX 2: Exam part 2 (CT2) – Assessment details and matrix

Component	Details
Exam part	CT2: Cumulative testing moment 2
Exam materials	All required literature and materials provided on Brightspace up until the date of the exam, including slides, exercises, and readings.
Format	6 multiple-choice questions and 4 open-ended questions.
Duration	60 minutes (conducted via Remindo).
Final grade weight	50% of the final course grade (max. 500 points).
Cut-off grade	Calculated after all three testing moments have been completed, and the exam is adjusted for difficulty.
Assessment dates	Consult your official timetable for the most up-to-date schedule. 1 st opportunity: Semester 1, December. 2 nd opportunity: Semester 2, March-April (as part of the full resit).

This exam part is the cornerstone of the course. It assesses your ability to conduct a preliminary case study analysis based on provided texts and then use that analysis to formulate a complete research design. The emphasis is on higher-order skills, with a focus on **Analysing** (i.e. conduct research on, distinguishes) information, **Evaluating** (i.e. appraising) research limitations, and **Creating** (i.e. formulate) your own research design.

Learning Goals assessed	Questions	CT2 weighting
2. The student conducts basic preliminary research on a safety and security topic using the 6W problem analysis framework on various types of literature.	6 MCQ (3-5 pts each)	25 pts = 50%
3. The student distinguishes the main concepts and variables of a safety and security study, as well as examines their connections.		
6. The student appraises safety and security studies in terms of potential limitations, such as data availability or access, sample size or representativeness, generalisability of results, time or budget constraints, and other relevant limitations.	1 open-ended question (6 pts)	6 pts = 12%
7. The student formulates their own applied research design, including a research problem, objective, central question, and two (2) sub-questions for a safety and security topic.	3 open-ended questions (5-7 pts each)	19 pts = 38%
Total:	10 questions (50 pts)	100%

APPENDIX 3: Exam part 3 (CT3) – Assessment details and matrix

Component	Details
Exam part	CT3: Cumulative testing moment 3
Exam materials	All required literature and materials provided on Brightspace for the entire course, including slides, exercises, and readings.
Format	20 multiple-choice questions.
Duration	30 minutes (conducted via Remindo).
Final grade weight	30% of the final course grade (max. 300 points).
Cut-off grade	Calculated after all three testing moments have been completed, and the exam is adjusted for difficulty.
Assessment dates	Consult your official timetable for the most up-to-date schedule. 1 st opportunity: Semester 1, February. 2 nd opportunity: Semester 2, March-April (as part of the full resit).

This final exam part assesses your comprehensive understanding of all course materials. It focuses on higher-level analytical and evaluative skills. While it covers concepts introduced after CT1, about a third of the questions will revisit earlier topics at a greater difficulty. The emphasis is on **Evaluating** (≈60%) and **Analysing** (≈20%) research, with the remainder on **Applying** (≈20%) concepts.

Learning Goals assessed	Questions	CT3 weighting
1. The student applies basic applied research concepts to different safety and security situations and contexts.	4-5 MCQ (1 pt each)	4-5 pts ≈ 13.3%
3. The student distinguishes the main concepts and variables of a safety and security study, as well as examines their connections.	4 MCQ (1 x 1 pt 2 x 2 pts 1 x 3 pts)	8 pts = 26.7%
4. The student evaluates different research questions, in terms of their type, clarity, focus, relevancy, feasibility, complexity, bias, and ethicality, within different safety and security contexts.	3-4 MCQ (3 x 1 pt 1 x 2 pts)	4-5 pts ≈ 16.7%
5. The student appraises the research design, including the research approach, research problem, objective, central question, and sub-questions of different safety and security studies.	3 MCQ (2 x 1 pt 1 x 5 pts)	7 pts = 23.3%
6. The student appraises safety and security studies in terms of potential limitations, such as data availability or access, sample size or representativeness, generalisability of results, time or budget constraints, and other relevant limitations.	5 MCQ (4 x 1 pt 1 x 2 pts)	6 pts = 20%
Total:	20 questions (30 pts)	100%

APPENDIX 4: Exam resit – Assessment details and matrix

Component	Details
Exam part	Cumulative exam resit
Exam materials	All required literature and materials provided on Brightspace for the entire course, including slides, exercises, and readings.
Format	46 multiple-choice questions and 4 open-ended questions. This single exam is a comprehensive assessment equivalent to CT1, CT2, and CT3 combined.
Duration	120 minutes (conducted via Remindo).
Final grade weight	100% of the final course grade (max. 1000 points).
Cut-off grade	Calculated after the exam is adjusted for difficulty.
Assessment dates	Consult your official timetable for the most up-to-date schedule. Semester 2, March-April.

The resit exam assesses all Learning Goals from the course and is an equivalent to CT1, CT2, and CT3 combined.

Learning Goals assessed	Questions	Resit weighting
1. The student applies basic applied research concepts to different safety and security situations and contexts.	19-20 MCQ (1 pt each)	19-20 pts ≈ 19-20%
2. The student conducts basic preliminary research on a safety and security topic using the 6W problem analysis framework on various types of literature.	6 MCQ (3-5 pts each)	25 pts = 25%
3. The student distinguishes the main concepts and variables of a safety and security study, as well as examines their connections.	4 MCQ (1 x 1 pt 2 x 2 pts 1 x 3 pts)	8 pts = 8%
4. The student evaluates different research questions, in terms of their type, clarity, focus, relevancy, feasibility, complexity, bias, and ethicality, within different safety and security contexts.	4-5 MCQ (4-5 x 1 pt 1 x 2 pts)	5-6 pts ≈ 5-6%
5. The student appraises the research design, including the research approach, research problem, objective, central question, and sub-questions of different safety and security studies.	7 MCQ (6 x 1 pt 1 x 5 pts)	11 pts = 11%
6. The student appraises safety and security studies in terms of potential limitations, such as data availability or access, sample size or representativeness, generalisability of results, time or budget constraints, and other relevant limitations.	1 open-ended question (6 pts)	6 pts = 6%
	5 MCQ (4 x 1 pt 1 x 2 pts)	6 pts = 6%
7. The student formulates their own applied research design, including a research problem, objective, central question, and two (2) sub-questions for a safety and security topic.	3 open-ended questions (5-7 pts)	19 pts = 19%
Total:	50 questions (100 pts)	100%

APPENDIX 5: Grading rubric for open-ended questions – CT2 and resit

Besides the MCQ, this exam includes an open-ended questions section. You will be given four short academic texts. Your task is to analyse these texts to **formulate a complete applied research design (~250 words total) for a specific client**. As was taught and practised in the course, your design must include the following four components:

- 1) Clear research objectives, with 2-3 (two to three) key actions (~60 words)
 - 2) One (1) central research question (max. 15-30 words)
 - 3) Two (2) sub-questions for the specific topic (max. 30-50 words)
 - 4) Two (2) potential research limitations (~80 words)
- **(5pts) Research objective**
 - (2pts) Focuses on an applied safety and security project for the client and their focus, connected to specific, relevant facts from the texts
 - (2pts) Clearly outlines 2-3 key actions you will take to answer your main research question, formulated according to SSMS standards (i.e. applied, focused, relevant, clear, feasible, complex enough, open-ended, unbiased in its use of language, ethical)
 - (1pt) Each key action starts with an action verb (e.g. to identify, to analyse, etc.)
 - **(7pts) Research question**
 - (1pt) Clearly aligned with the client's focus, connected to specific, relevant facts from the texts
 - (1pt) Clearly aligned with the stated research objective (i.e. the key actions)
 - (5pts) Formulated according to SSMS standards (i.e. formatted as a question with a single topic, applied, focused, relevant, clear, feasible, complex enough, open-ended, unbiased in its use of language, ethical)
 - **(7pts) Research sub-questions**
 - (3pts, 1.5pt per question) Includes two (2) distinct questions that both directly help answer the central research question
 - (4pts, 2pts per question) Two questions (2) questions formulated according to SSMS standards (i.e. formatted as a question with a single topic, applied, focused, relevant, clear, feasible, complex enough, open-ended, unbiased in its use of language, ethical)
 - **(6pts) Research limitations**
 - (2pts, 1pt per limitation) Identifies two (2) distinct and relevant limitations for the proposed research (i.e. aligned with the research objective, the central research question, and/or one of the sub-questions)
 - (4pts, 2pts per justification) Provides a clear justification for each limitation, explaining its potential impact on the proposed research, based on specific, logical arguments.

For your applied research design, you must base your answers only on the information found in the provided texts.

Your justifications of the research limitations can include logical arguments that go beyond the provided texts, as long as they are relevant to your design.

APPENDIX 6: Grading rubric for quizzes and quizzes resits

The three quizzes are assessed via Brightspace, using a Pass or Fail rubric. Throughout the semester, these quizzes prepare you directly for the different exam moments in the course:

- Quiz 1 focuses on part 1, Introduction to Research Methods, which is tested in CT1 and CT3.
- Quiz 2 focuses on part 2, Understanding the Problem, which is tested in CT2.
- Quiz 3 focuses on part 3, Planning your Investigation, which is also tested in CT2.

The questions in Quiz 1 follow the [assessment details and matrix of CT1](#), but note that only the concepts in part 1 of the concept list are included in Quiz 1 and its resit, and that the assignment will be graded as Pass or Fail.

The questions in Quiz 2 and Quiz 3, together, serve as an example of what CT2 will look like, and thus follow the [assessment details and matrix of CT2](#). These quizzes, and their resit, are also graded as Pass or Fail.

You must receive a Pass for the three quizzes, or their resits. You are encouraged to use any remaining quizzes as preparation for the exams.

APPENDIX 7: Concept list

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Part 1: Intro to Research Methods

Informal approach

Explanation: Using an informal approach means drawing conclusions based on observations in daily life (e.g. what you see and your opinion) rather than using standard research methods. It is a quick, easy, and flexible way to make decisions that works well for most everyday situations. However, because it relies on your personal frame of reference, it is vulnerable to bias and subjectivity. For this reason, a systematic approach is recommended in high-stakes situations.

Example(s): You want to form a study group for an upcoming exam. You might approach the classmates you sit next to, send a message in a group chat, or observe who seems most engaged in class. While these strategies are quick and will likely work, they are vulnerable to bias. For instance, you might only talk to people you already know but are not compatible as study mates (e.g. different learning styles or grade ambitions), or you might exclude shy but knowledgeable students.

Additional remarks and resources:

- ☐ Read: Verhoeven, 2019, p.22 (section 1.1 in HHS online version).
- ☐ Note: Most documentaries use informal approaches. They do not follow a systematic approach, featuring informal talks with a few selected people giving their personal opinion. This can be entertaining, but it is not an acceptable approach when the results will influence people's safety. Always consider, what type of research is it based on?

Systematic approach

Explanation: A systematic approach refers to observations that are based on adequate research, following standard research methods according to a fixed, methodological plan. It is a intentional set of organised activities designed to reach an objective. This approach is essential in professional and high-stakes situations where unbiased and reliable conclusions are necessary.

Example(s): You want to find out how many students are aware of the emergency exits on campus. To ensure your study is reliable, you design a structured survey, using standard research methods practices. This survey is a standardised way to ask each student the exact same set of clear and concise questions to avoid different interpretations.

Additional remarks and resources:

- ☐ Read¹: Verhoeven, 2019, p.22, online 1.1.
- ☐ Note: All academic articles used in this course follow a systematic approach to research and are considered reliable sources of information.

¹ All readings can be found on Brightspace or via the THUAS/HHS Library (e.g. online section 1.1.).

Research

Explanation: Research is a systematic (and as far as possible) approach involving an organised investigation into a topic to establish facts and draw conclusions. It is based on the [scientific method](#), a cyclical process of making observations, asking questions, conducting strict research, and sharing the results. In applied research, you begin this cyclical process by identifying a need (i.e. making observations), establishing a theoretical foundation (i.e. understanding the “why” based on existing research), developing a research question, and designing the study (i.e. creating your own applied research design). You will learn about these first steps in this course. In the second semester, you will learn more about the other steps in the cyclical process.

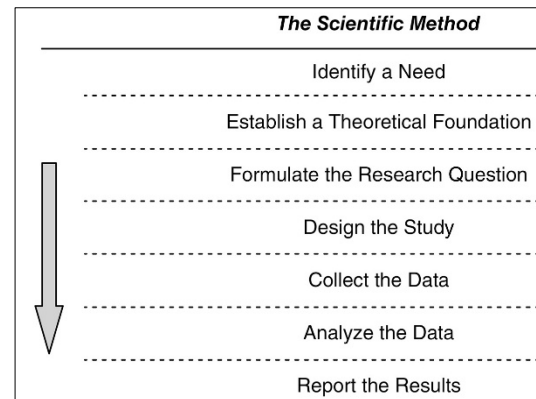


Figure 1. *The Scientific Method*. From *A Primer of the Scientific Method and Relevant Components*, by Edmonds & Kennedy, 2017.

Example(s): A researcher systematically analyses data about the causes of traffic accidents near schools. Based on a literature review (i.e. reading and analysing a lot of academic texts containing research findings), the researcher recommends implementing different road designs to improve safety.

An organisation conducts an investigation into a series of data breaches. The research involves studying server logs and interviewing IT staff to establish the facts about how the breach occurred and draw conclusions on how to prevent it in the future.

Additional remarks and resources:

- ☐ Watch: [To hear this explained in different words, watch "What is research?" \(3:55\).](#)
- ☐ Note: An investigation is a formal inquiry or examination of a particular subject; however, not all investigations use proper research methods. For an investigation to qualify as research, it must be based on the scientific method.

Fundamental research & theoretical application

Explanation: Fundamental research, also known as pure or basic research, is used to explore topics and build upon or develop new theories and hypotheses. The primary aim of this type of research is to produce and improve knowledge for its own sake. It focuses on understanding basic principles.

Example(s): A study that aims to increase our understanding of how risk management decisions are affected by cognitive biases, such as the availability bias. To do this, researchers conduct simulations with participants in controlled laboratory settings.

A project focused on developing a new method for evaluating fear based on various measurements, such as differences in facial temperature.

Additional remarks and resources:

- Read: Verhoeven, 2019, pp. 23-24, online 1.2.
- Watch: [To see an example of fundamental research, watch "Two views on Human Error" \(5:52\). After 03:50, the "system improvements" mentioned are where applied research begins.](#)

Applied research & practical application

Explanation: Like fundamental research, applied research is conducted to acquire new knowledge. However, findings of applied research are primarily aimed at resolving a specific, practical problem. Because of this, the results of applied research are usually intended to be primarily valid for a single or limited number of situations with a similar context. Applied research can be a follow-up to the findings of fundamental research (Verhoeven, 2019, pp.23-24; online 1.2).

Example(s): A project investigating how to improve communication between employees of a lab in Delft and their onsite safety officer.

A project investigating the safety perception of refugees living in Rijswijk.

Additional remarks and resources:

- Read: Verhoeven, 2019, pp.23-24, online 1.2.
- Watch: [To see an example of applied research, this video shows you the applied side of theoretical perspectives on human error, in this case, applied to a process plant scenario \(3:56\).](#)

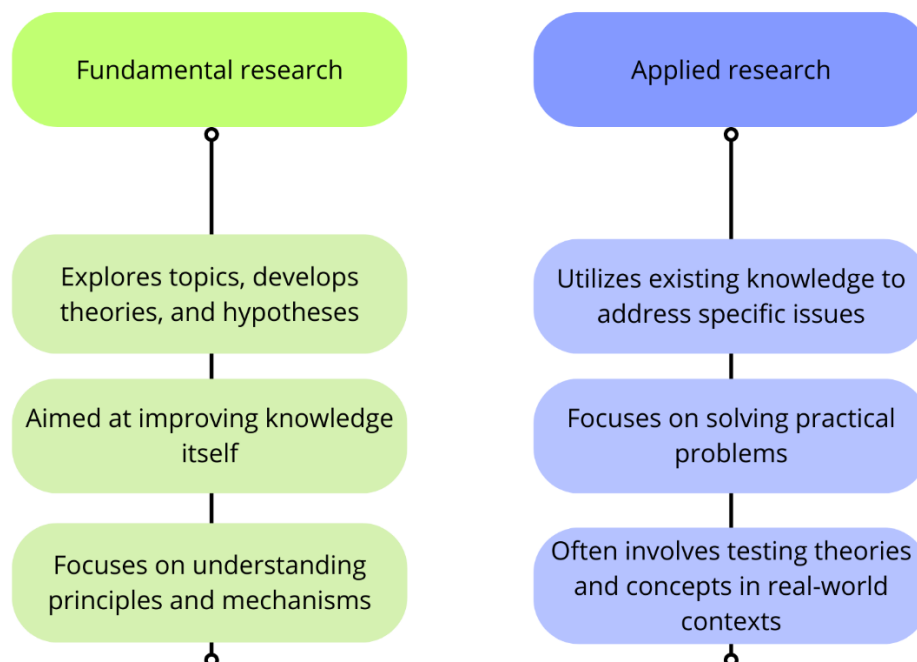


Figure 2. Fundamental research vs applied research.

Research methods

Explanation: Research methods are the specific procedures, strategies, and techniques used for systematically collecting and analysing data (for both fundamental and applied research). These methods are the tools that allow researchers to uncover new information or create a better understanding of a topic. In SSMS, you will write a methodology section for your reports and thesis, which requires you to explain which research methods you used and how you used them.

Example(s): The scientific method is a foundational research method that follows five basic steps: observation, question, hypothesis, prediction, and test. In your next research methods course, you will learn about specific methods like designing surveys or conducting structured interviews to collect data.

Additional remarks and resources:

- ☐ Read: Verhoeven, 2019, p.23-24; online 1.2.
- ☐ Explore further: [Learn more about the scientific method that guides research.](#)

Research problem

Explanation: Developing a research problem helps you understand what the problem is, how it affects social, how all its different elements interact, and what you as a researcher can focus on. It is a statement about an area of concern, a knowledge gap, or a situation that needs to be improved. It points out the need for an investigation and helps you set boundaries for your research topic. A well-defined research problem is the foundation for designing a focused and manageable research objective and research question.

Example(s): Schiphol Airport authorities are under pressure to enhance security measures due to the evolving tactics of potential threats. There is a pressing need to investigate vulnerabilities in existing airport security protocols in light of emerging threats like new forms of contraband, cyber-physical risks, and insider collusion.

Adobe, a multinational corporation has experienced a series of data breaches, raising concerns about insider threats. A critical knowledge gap exists within large organisations regarding the identification, assessment, and mitigation of insider threats.

Additional remarks and resources:

- ☐ Read: van Tulder, 2018, pp.88-90; Verhoeven, 2019, pp.55-59 (online 3.1-3.2.1).
- ☐ This concept will be discussed in-depth in lecture 3 and practiced in workshop 3.

Research objective

Explanation: Sometimes used interchangeably with research goals, for the purpose of SSMS, this is a statement or two that summaries what you are trying to achieve with your research. It should be practical and action-oriented, serving as a direction that helps you stay focused on a clear goal. Your research objective is a key part of your "[golden thread](#)", ensuring that every part of your project is

logical and consistent. For this reason, it also needs to align with your research question(s). Consider using the [SMART model](#) (Specific, Measurable, Achievable, Relevant, and Timely/Timebound) when developing your research objective.

Example(s): By the end of this six-month project, this research aims to identify and evaluate at least three key vulnerabilities in the current security protocols at Schiphol Airport and deliver a report with evidence-based recommendations for improvement. First, a survey will be conducted...

This research seeks to analyse the nature of insider threats at Adobe and develop a set of at least five targeted prevention strategies, to be presented to the management board by December 1st. First, a literature review will be conducted...

Additional remarks and resources:

- ☐ Verhoeven, 2019, pp.73-83 (online 3.2.3).
- ☐ This concept will be discussed in-depth in lecture 3 and practiced in workshop 3.
- ☐ Explore further: [Writing SMART objectives.](#)

Research question

Explanation: A research question (RQ), also known as the central or main question, is the single question that your entire research project is designed to answer. It is the driving force behind your research and should guide every decision you make. The results of your research should directly answer this question.

Example(s): How can security at Schiphol airport be improved to protect against new threats like cyber-physical risks?

How can effective strategies for early detection and prevention of insider threats be developed at Adobe?

Additional remarks and resources:

- ☐ Read: van Tulder, 2018, pp.88-90; Verhoeven, 2019, pp.73-83 (online 3.1-3.2.3).
- ☐ This concept will be discussed in-depth in lecture 3 and practiced in workshop 3.

Research sub-questions

Explanation: Research sub-questions (SQs) are secondary questions that help you explore different aspects of your main research question. They break down a complex topic into smaller, more manageable parts, which helps to structure your research. They must all be related and contained within the limits of the main research question. If you have a very complex topic, you can divide it into as many sub-questions as necessary (Verhoeven, 2019, pp.73-83; online 3.1-3.2.3). For SSMS projects, you will usually not use more than 3-5.

Example(s): RQ: How can security at Schiphol Airport be improved to protect against new threats like cyber-physical risks?

- SQ1: What are the current security measures at Schiphol Airport against cyber-physical risks?

- SQ2: What specific measures against cyber-physical risks have been successful in other large international airports, which could be applied to Schiphol Airport?

RQ: How can effective strategies for early detection and prevention of insider threats be developed at Adobe?

- SQ1: What are the key technological and behavioural factors contributing to insider threats within Adobe?
- SQ2: What specific strategies and technologies, proven effective in similar corporations, can be adapted to enhance cybersecurity and data protection at Adobe?

Additional remarks and resources:

- ☐ Read: van Tulder, 2018, pp.88-90; Verhoeven, 2019, pp.73-83 (online 3.1-3.2.3).
- ☐ This concept will be discussed in-depth in lecture 3 and practiced in workshop 3.

Research ethics

Explanation: Research ethics refers to the ethical principles that guide all research. In the field of safety and security, research often involves people, so it is crucial that your work is conducted according to high ethical standards. In this course, the Netherlands Code of Conduct for Research Integrity (2018, pp.13-14) will be used as guidelines. The five core principles are:

- **Honesty:** Being truthful in your work, using reliable data (e.g. from academic articles and other reputable sources), and avoiding plagiarism by properly citing all your sources.
- **Scrupulousness:** Using appropriate scientific methods and designing your research carefully.
- **Transparency:** Being clear about what data you used, how you obtained it (i.e. which methods you used and how you used them), and what your results are, so that your work is verifiable. Transparency also requires full disclosure and openness regarding potential conflicts of interest and external support (e.g. funding sources, financial support, agreements with externals regarding rights to data access, etc.), so that other researchers and the public can assess the research's objectivity.
- **Independence:** Ensuring that your research process is guided by scientific considerations, not outside pressures (e.g. non-scientific considerations, external financial/commercial interests, clients, funders, other external stakeholders). While your choice of research topic and research question (i.e. the problem to investigate) can be guided by what is required in society, your design and research process must be independent. Research funding conditions or agreements with external sponsors must never allow the funder to dictate, manipulate, or block methods, influence research findings, or suppress results; these are external influences that are very challenging to manage.
- **Responsibility:** Conducting research that is relevant to society and always considering (within reason) the interests and safety of your research participants, as well as other stakeholders involved. This can also include the informed consent (participants should be aware of what data you intend to collect and what you will do with it, avoid deceiving or misleading them) and protection of participants (they should be protected from physical or mental harm).

Example(s): Plagiarism is a violation of the principle of *Honesty*. It includes directly copying someone else's work, paraphrasing to disguise copying, or providing inaccurate citations that make a source

impossible to find. Citing a source with inaccurate or incomplete information, making it impossible to trace that source back to its origin, is also considered plagiarism.

The Stanford Prison Experiment (1971) is a famous example of unethical (and unreliable²) research. This Navy-funded experiment examined the effects of power dynamics between prison officers and prisoners. Participants who were assigned the role of "guards" began to psychologically and physically harm the "prisoners", who eventually also harmed the "guards". This represented a serious failure of the principle of *Responsibility*, specifically the duty to protect participants from harm.

Additional remarks and resources:

- ☐ Read: [Netherlands Code of Conduct for Research Integrity, 2018, pp.13-14.](#)
- ☐ Watch: [To learn more about ethical principles, watch "Ethical Principles in Research Involving Humans as Subjects" \(8:21\).](#)

Data

Explanation: A collection of raw facts, which on their own may not have a purpose and require processing. The data points that make up data can be quantitative (e.g. numerical, statistical) or qualitative (e.g. descriptive, such as texts or images). It may seem useless until it is organised and interpreted to give it meaning (i.e., processed and turned into information).

Examples:

- 2012, 1,127,900; 2022, 797,700 (Statista Research Department, 2023).
- In 2021, the Dutch police received a score of 6.3 out of 10 in terms of how much trust Dutch citizens had in them (Statista Research Department, 2023).

Additional remarks and resources:

- ☐ Read: [To learn more, you can read about the difference between data and information here.](#)

Information

Explanation: Information is the result of processing (i.e. organising and interpreting) raw data. It is the perception of data, gained through analysis, which puts a set of data points into a context that has meaning.

Example(s): Between 2012 and 2022, the number of registered crime cases in the Netherlands decreased from 1,127,900 to 797,700 (Statista Research Department, 2023).

² Despite their popularity, famous psychological studies are often fraudulent or outdated. These studies usually do not follow ethical standards, lack concrete data, exaggerate their results, and come to incorrect conclusions. They are also not replicable by other researchers, which suggests that their process was not transparent (i.e. they are not honest about how obtained these results). In the case of the Stanford Prison Experiment, there is [evidence that, among other things, both the "guards" and the "inmates" knew the results the researchers wanted](#) from them and tried to meet those goals, which is an example of social desirability bias.

The score of 6.3 out of 10 for how much Dutch citizens trust the police is a 0.4-point increase compared to 2012 and 2013 (Statista Research Department, 2023).

Additional remarks and resources:

- Watch: [To learn more, you can watch a video about the difference between data and information here \(2:56\).](#)

Fact (empirical evidence)

Explanation:

A fact is information that is demonstrably true and can be proven through research methods, such as observation or measurement. A factual claim should be traceable to reliable evidence and open to checking. Not all facts are universal; many are time- and context-specific, such as recorded crime figures. Facts are used as empirical evidence in research to support claims, generalisations, and conclusions. To test whether something is a fact, ask yourself:

- Can it be applied to all similar situations?
- Can it be duplicated with the same results?
- Can it be verified by primary sources?
- Has it been historically recorded?
- Can it be observed?

Examples: "Water boils at 100°C at sea level." This is a fact because it can be consistently observed, measured, and duplicated in experiments under standard atmospheric conditions. This has been extensively reported in scientific literature.

"The total number of registered crime cases in the Netherlands in 2022 was 797,700." This is a fact because it is a piece of historical data that can be verified through official sources like Statista. Nevertheless, always check the original source where possible.

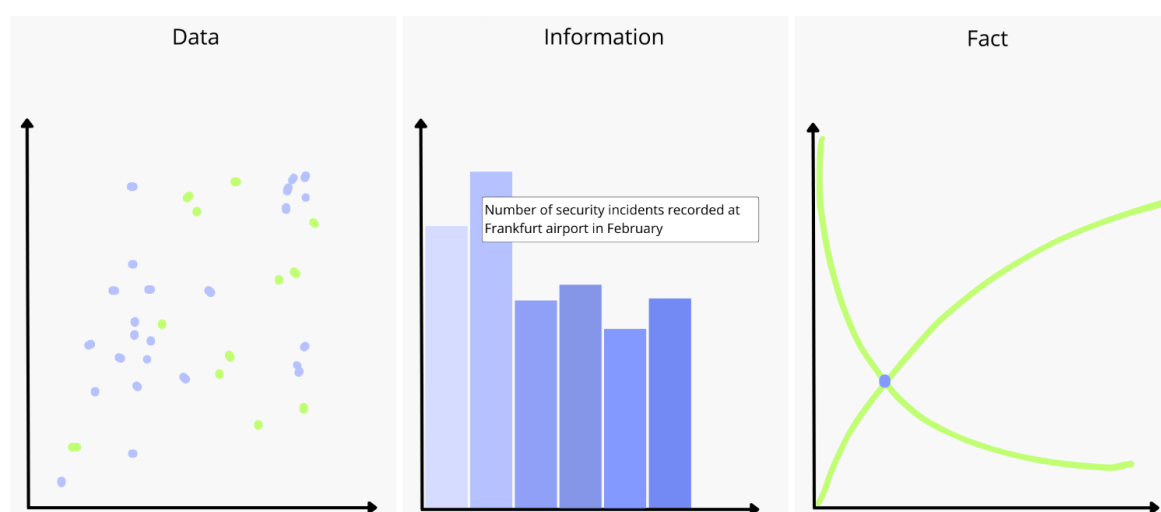


Figure 3. Information, data, fact (i.e. something proven to be true)

Additional remarks and resources: Watch: [To learn more about the difference between facts and opinions, watch "Fact or opinion" \(4:53\).](#)

Theory

Explanation: In daily life, a "theory" is often assumed to be a guess. In research, however, a theory is a logically consistent explanation about a specific behaviour or set of events that is based on known principles and is validated by empirical evidence. It provides an explanation for understanding the process and tries to predict what will happen in similar situations. Theories can change as more research is conducted.

Example(s): Critical theory (1930s) is a sociological perspective that focuses on understanding and challenging power dynamics and inequality in society (many authors have contributed to this theory, such as Herbert Marcuse and Max Horkheimer). It analyses existing social structures, the way power is unevenly distributed, and the way it is used to maintain and reproduce social inequalities. It seeks to uncover and question the assumptions that lead to oppressive systems and social injustice.

Sociology is the study of social life and social change, so they are often used in SSMS. [You can find other examples of sociological theories used in safety and security in this video \(06:30\).](#)

Additional remarks and resources:

- Watch: [To learn more, watch "Why science is NOT 'Just a Theory'" \(3:43\).](#)

Framework

Explanation: In research, a framework is a structure, model, or tool used to investigate a research problem. It offers a specific, often step-by-step, approach to analysing information or solving a problem. Frameworks are used as practical tools to help researchers or practitioners work on complex issues; it often complements (or is complemented by) a theory.

Example(s): A SWOT analysis is a framework that individuals or organizations can use to analyse their Strengths, Weaknesses, Opportunities, and Threats. This tool provides a structured way to examine these factors and can help challenge assumptions and uncover dangerous blind spots.

A Risk Matrix is a framework that provides a structure to analyse the probability and impact of different risks. By providing an overview of the severity of various risks, this tool helps you prioritise which ones to address first. You will learn more about this tool in later SSMS courses.

Additional remarks and resources:

- Read: [Maslow's hierarchy of needs is a model based on psychological theories of motivation. It consists of a five-tier structure of hierarchical human needs. You can read more about it here.](#)

Opinion

Explanation: An opinion is a personal view or belief about a topic, which is not necessarily based on or supported by empirical evidence (factual information).

Example(s): "I believe that safety and security are important for society." (This is an opinion that can be supported by empirical evidence)

"I believe that birds are not real but are actually government surveillance drones." (This is an opinion not based on empirical evidence).

Additional remarks and resources:

- ☐ Watch: [To learn more about the difference between facts and opinions, watch "The Blur Between Facts and Opinions in the Media" \(3:13\).](#)

Cognitive biases

Explanation: A cognitive bias is a pattern of systematic errors in thinking that results in a misperception of reality. These errors happen unconsciously due to the way our brains process information and can make us vulnerable to manipulation. To succeed in the field of safety and security, you must learn to recognise these biases in yourself and others. This will help you to avoid being manipulated by salespeople, politicians, and other skilled manipulators. There are many cognitive biases, but course will focus on five key biases: confirmation bias, availability heuristic, cognitive dissonance, social desirability, and the Dunning-Kruger effect.

Example(s): Humans have the tendency to estimate the likelihood of an event based on how easily and readily they remember examples of it. If something is easily remembered, people tend to believe it is more common or likely to happen, even if that is not necessarily the case. This is why a vivid news report about a rare event can make it seem more prevalent than it actually is. This is called the availability heuristic, and it contributes to our frequent failure to notice the vast amount of evidence ignored by a debater "cherry-picking" (a common logical fallacy) their points.

Social desirability bias is the tendency for people to answer questions in a way they believe will be viewed favourably by others. For example, if you are conducting a survey to assess employees' compliance with security protocols, employees might overstate their adherence to the rules because they know that is the "correct" answer and want to be seen in a positive light. In other words, a participant might feel inclined to tell you what they think you want to hear in that situation, rather than their real, unfiltered opinion. To reduce this effect, the anonymisation of all participant data is recommended.

Additional remarks and resources:

- ☐ Read: [To learn more, you can read more about cognitive biases here.](#)
- ☐ Note: You will have to research the other key biases yourself.

Logical fallacies

Explanation: Errors in reasoning that are related to how you make claims and construct arguments. Like cognitive biases, logical fallacies often occur unconsciously, but unlike cognitive biases, they might also be committed intentionally to persuade and mislead others. For instance, logical fallacies can be used during speeches, disagreements, and other discussions with the intention of using erroneous arguments to make a point, or win an argument. This course will focus on ten key fallacies: anecdotal

(aka hasty generalisation), appeal to authority, causal, cherry picking, ambiguity (aka equivocation), personal incredulity, burden of proof, slippery slope, strawman, and ad hominem.

Example(s): An anecdotal fallacy occurs when someone uses a personal experience or an isolated example to dismiss statistics or other evidence. For instance: "I do not need to wear a helmet when I ride my motorcycle. My father has ridden a motorcycle for decades without one and nothing bad ever happened to him." This is also called anecdotal evidence.

- Personal incredulity becomes a fallacy when someone assumes that information must be false because they do not personally understand it or are unable to fully grasp some of the details. For instance, "I do not understand how vaccines work, therefore vaccines do not work". A person's inability to comprehend a concept does not impact its veracity.
- The strawman fallacy happens when someone misrepresents or distorts an opponent's argument to make it easier to attack. For example, if someone argues for stricter environmental regulations, a person using a strawman might respond, "So you want to shut down all factories and destroy the economy?" This is not what was originally argued.

Additional remarks and resources:

- Read: [To learn more, you can read about logical fallacies here](#)

Qualitative research

Explanation: Qualitative research focuses on gathering non-numerical data about lived experiences, emotions, or behaviours, and the meanings individuals attach to them. It helps researchers gain a deep understanding of complex concepts, social interactions, or cultural phenomena. This type of research is useful for exploring how or why things have occurred by interpreting events and describing actions, often through methods like interviews and literature reviews. It focuses on generating meaning through rich descriptions. The majority of the research conducted in SSMS is of a qualitative nature.

Example(s): You are working on a project about crisis management during wildfire season and want to understand the perspectives of crisis managers. You could conduct in-depth interviews or observations to collect qualitative data on their subjective experiences and ideas.

A team of researchers is studying past pandemic responses to offer recommendations for the future. They conduct a literature review, analysing academic articles and government reports to produce recommendations based on existing qualitative information.

Additional remarks and resources:

- Watch: [To learn more, watch "Quantitative vs. Qualitative Research: The Differences Explained" \(3:36\).](#)

Quantitative research

Explanation: Quantitative research focuses on gathering numerical and statistical data that can be ranked, measured, or categorised through statistical analysis. It is used to uncover patterns or

relationships and to make generalisations. This type of research is useful for finding out how many, how much, how often, or to what exact extent something occurs, often through methods like surveys and questionnaires.

Example(s): A government agency conducts a quantitative study to assess the number of cyberattacks on critical infrastructure over the past year. They collect a large amount of numerical data from companies throughout the country using a large-scale survey.

In a quantitative study, a security company analysed data from 500 cyberattacks to assess the impact of response time on data breaches. Results showed that faster incident detection reduced breach severity by 25%, emphasising the importance of quick response in cybersecurity.

Additional remarks and resources:

- Watch: [To learn more \(and practise with exercises\), watch "Empirical Studies: Qualitative vs. Quantitative" \(5:51\).](#)

Reading techniques

Explanation: There are many reading techniques out there. At SSMS, we advise three: scanning, skimming, and detail reading. Think of it like looking at a city map:

- Skimming is glancing at the whole map to see the general layout of the city.
- Scanning is looking for a specific street name.
- Detail reading is tracing the exact route from one point of the city to another.

Scanning is used to quickly decide whether the text you are reading has what you need. Skimming is used to rapidly get an overview of the entire text. You go for the "spoilers", making sure to prioritise introductions and conclusions, reading only the first/last paragraphs of each. This helps you narrow down the parts of the text that you need to take a closer look at. Finally, detail reading is what you do when you have found the parts of the text you need to carefully read and study.

Techniques	Meaning	Uses	Tips
Skimming	Reading a text quickly for a general overview	To get a rough idea of what the text is about	• Read headings • First and last sentences of paragraphs (topic and concluding sentences)
Scanning	Searching for specific information	To locate a specific fact, date, or keyword	• Look for tables, graphs • Look for specific words (names, cities, countries etc.)
Detail reading	Reading a passage as carefully as you can	To understand every detail in the passage	• Read every word • Pay attention to grammar and punctuation • Notice linking words and phrases

Critical reading	Analysing and evaluating the text's arguments and evidence.	Goes beyond simply location a specific piece of information and understanding it.	• Read a lot and practice. • Limit your AI usage when interpreting texts. • You will learn how to do this throughout your time in SSMS.
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Example(s):

Additional remarks and resources:

- ☐ Read more information about these techniques in the slides of workshop 3.

Deductive thinking

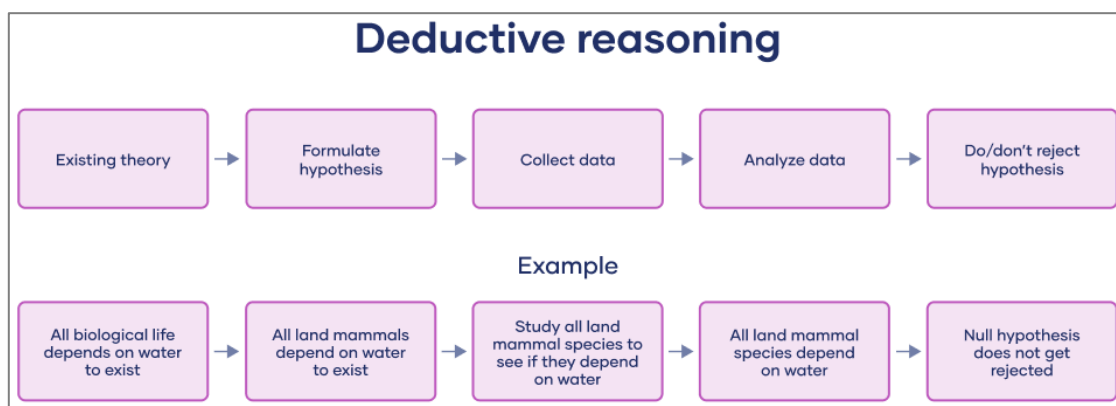
Explanation: A systematic, top-down approach that starts with a general theory or rule and moves to a specific conclusion. If the general premises (. i.e. a statement from which another follows as a conclusion, see examples below) are true, the conclusion must also be true. This approach is strongly associated with quantitative research, where a hypothesis (derived from a theory) is tested with large amounts of data.

Example(s):

- Premise 1: All birds have wings (general principle or fact).
Premise 2: A crow is a bird (specific case).
Conclusion: A crow has wings (more specific conclusion).
- Premise 1: All unencrypted network traffic is vulnerable to interception (general principle).
Premise 2: The data being sent to Server X is unencrypted (specific case).
Conclusion: Therefore, the data being sent to Server X is vulnerable to interception (more specific conclusion).

Additional remarks and resources:

- ☐ Watch: [To learn more, watch "Introduction to Inductive and Deductive Reasoning" \(03:33\).](#)



- Figure 4. What is deductive reasoning? *Explanation and examples available via Scribbr*

Inductive thinking

Explanation: A systematic, bottom-up approach that starts with specific observations (also expressed in premises) and moves to find more general patterns, conclusions or theories. This generalised conclusion is broader than the limited data or information you have to back it up. This means that the conclusion is likely based on the evidence, but it is not guaranteed to be true once more data or information is uncovered. This approach is often used in exploratory research to generate new hypotheses and begin working towards proven theories. Because of this, it is often an approach that occurs before deductive research takes place.

Examples(s):

- Premise 1: A crow has wings (limited observation from a specific case).
Premise 2: A crow is a bird (specific case).
Conclusion: All birds have wings (plausible conclusion about a larger set of cases, but you need more data than that provided by crows to back it up).
- Premise 1: Computer A in the finance department is infected with ransomware. Analysis shows the infection started with a fake invoice PDF opened by an employee (limited observation from a specific case).
Premise 2: Computer B in the marketing department is infected with the same ransomware. The infection also started with a fake invoice PDF (specific case).
Conclusion: A widespread phishing campaign using fake invoices as bait is currently targeting our company (conclusion is logically sound, considering the data you have, but might not be true, i.e. the infections could have different origins, but happened to occur around the same time those attachments were opened).

Additional remarks and resources:

- Read: [To learn more, read about the differences between inductive and deductive reasoning here \(with more examples\).](#)

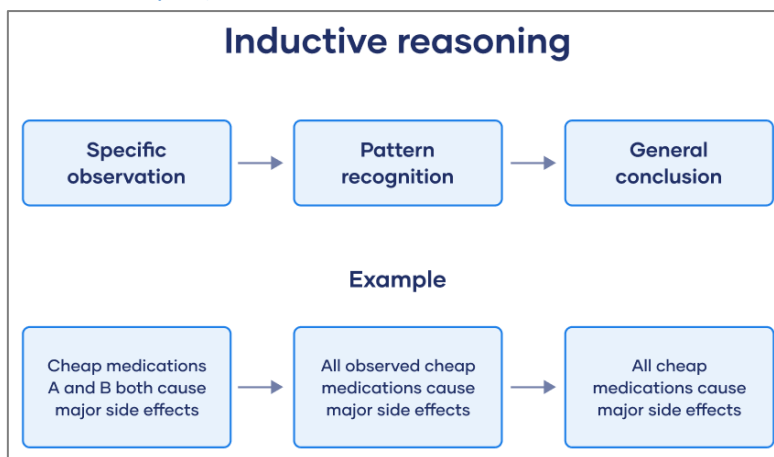


Figure 5. What is inductive reasoning? [Explanation and examples available via Scribbr](#)

Mixed-methods research

Explanation: A research approach that combines quantitative and qualitative methods to gather and analyse data. This provides a more holistic understanding by integrating statistical data with deeper contextual insights. For example, a project might start with qualitative interviews to develop a hypothesis and then use a quantitative survey to test it on a larger population.

Example(s): To evaluate school security, you use a survey (quantitative) to gather student perceptions and conduct interviews (qualitative) with security staff to understand their challenges.

Additional remarks and resources:

- ☐ You will learn more about the different types of research methods in your next research methods course: ART.

Multi-method research

Explanation: A research approach that uses two or more methods from the same family—either multiple qualitative methods or multiple quantitative methods—in a single study. This is different from mixed-methods research, which combines qualitative and quantitative approaches. Multi-method research is used to gain a deeper, more comprehensive understanding of a topic from different angles within the same research paradigm. This is like using two different types of screwdrivers (a Phillips head and a flathead) to build a cabinet. You're using multiple tools, but they are from the same family (screwdrivers).

Example(s): To study workplace safety culture, a researcher conducts in-depth interviews with employees and also performs direct observations of their daily routines. Both methods are qualitative

To analyse the effectiveness of a new security system, a researcher sends out a survey to gather user satisfaction data and also analyses system log data to measure the frequency of security alerts. Both methods are quantitative.

Additional remarks and resources:

- ☐ You will learn more about the different types of research methods in your next research methods course: ART.

Triangulation

Explanation: Triangulation means using multiple viewpoints to make your findings more reliable—like looking at an object from different angles to see it more clearly. This helps the researchers confirm the findings by cross-verifying the results through various standardised, systematic methods. There are two main types:

- Data triangulation: Using different sources of data (e.g. interviewing students, teachers, and parents on the same topic of classroom safety).

- Methodological triangulation: Using different methods to study the same problem (e.g. using both surveys and observations to look into how safety is discussed in the classroom).

Mixed-methods research is one powerful way to achieve methodological triangulation, but you can also triangulate using multi-method research with multiple qualitative methods (like interviews and focus groups).

Example(s): Data triangulation: To validate information from one study, you compare it with several other studies on the same topic.

Methodological triangulation (using only qualitative methods): You analyse existing studies on a topic and also conduct interviews with experts to get more nuanced information.

Additional remarks and resources:

- Watch: [To learn more about triangulation, watch this video "Triangulation: Combine Findings from Multiple User Research Methods" \(02:10\)](#)

Holism

Explanation: Holism means understanding that a system (like workplace safety or an ecosystem) is more than just the sum of its parts. To understand it, you must look at the whole picture and the connections between the parts. This approach helps you manage a situation more comprehensively by taking all related factors into account.

Example(s): Holistic workplace safety considers all factors that contribute to the employee's physical (e.g. safety equipment, hazard identification), emotional (e.g. work-life balance, conflict resolution), and mental wellbeing (e.g. stress management, mental health awareness).

An ecosystem is more complex than the sum of its living organisms, as it also includes inorganic elements such as soil and water, and the interactions between all these elements. Safety and security research topics operate the same way. Keep an eye out for all the factors related to your topic (including, very often, the ecosystem it is located in).

Additional remarks and resources:

- Watch: [Check out "A Holistic Approach to Urban Security and Management" for a very brief example about urban security \(01:33\).](#)

Part 2: Understanding the problem

Research project

Explanation: A scientific effort to answer a research question within a limited time, following a structured approach to achieve an observable result. A research project can be applied (solving a practical problem) or fundamental (developing a theory) and can use qualitative, quantitative, mixed-methods, or multi-methods. The approach you choose depends on your research goal. The reasons behind research projects can range from wanting to answer a practical question by gathering information (e.g. exploratory or descriptive research, like we do in applied sciences universities) and/or the need for fundamental research focused on theory testing or developing (e.g. explanatory or predictive research, like what is done at fundamental research universities).

Example(s): A systematic investigation into the link between money laundering and organised crime in Brazil.

A systematic study into the potential challenges that weather-related hazards, such as droughts, floods, and hurricanes might pose to nuclear power plants.

Additional remarks and resources:

- ☐ All projects in SSMS are considered research projects. This course prepares you for your first project in semester 2 by practicing the first steps of all research projects (preliminary research and developing an applied research design).

Preliminary research

Explanation: This is the initial phase of any research project, where you find basic information about your topic, answering questions such as who, what, when and where. This initial research will also help you determine controversies related to your topic and determine if there are enough sources available to cover the topic effectively. It is often recommended to follow a framework (e.g. 6W) while conducting this type of research, to obtain a rich overview of the topic.

Example(s): For a paper on the 2020 California wildfires, your preliminary research would be a quick check on the causes, affected areas, and timeline. Your background research would be a deep dive into existing studies on wildfire management, climate factors, and community impact.

Additional remarks and resources:

- ☐ Read: van Tulder, 2018, pp.88-90; Verhoeven, 2019, pp.55-59 (online 3.1).

Background research

Explanation: Following the preliminary research phase, once you have a solid research design (with a defined and clear research objective, research problem, research question, and sub-questions), you can conduct extensive literature research. You do this to get more familiar with the information that

is already out there about the problem, before you start collecting and analysing your own data for a research project. You will learn how to do this in the second semester, in the ART and the Research Project courses.

Examples(s): You are writing a paper about the Somali civil war, after conducting your preliminary research you do a more comprehensive and in-depth examination of the topic. Its purpose is to develop a thorough understanding of the subject matter.

Additional remarks and resources:

- ☐ Read: van Tulder, 2018, pp.88-90; Verhoeven, 2019, pp.55-59 (online 3.1).

6W/5W1H problem analysis framework

Explanation: A structured approach used to analyse a problem from six key perspectives: what, why, where, who, when, and which (or how). Looking at a situation from these different angles helps you gain a comprehensive understanding of the problem. Thus, this is a problem analysis framework.

Example(s): See any CT2 exercise answer key offered in this course.

Additional remarks and resources:

- ☐ Read: van Tulder, 2018, pp.88-90; Verhoeven, 2019, pp.55-59 (online 3.1).
- ☐ Watch: [To see this explained in other words, watch "How to Question with 5W1H" \(03:18\).](#)

Research proposal

Explanation: A structured, formal document that explains what you plan to research (your research design), why it is worth researching (its relevance), and how you will conduct the research (your methodology). In this course, you practice creating the research design (i.e. research problem, research objective, research question, and sub-questions) and justifying its relevance.

Example(s): SSMS students work hard on their research proposals because this proposal needs to be approved before they can move on to their final project, the thesis.

A researcher prepares a research proposal to apply for project funding.

Additional remarks and resources:

- ☐ Later in the programme, once you have a solid grasp of all parts of the process, you will have to write a full research proposal for your thesis.

Case study

Explanation: A detailed investigation of a particular problem, person, group, or event. It is like zooming in on one piece of the puzzle to understand how it fits into the bigger picture. Researchers collect detailed information, often from the subjects themselves (e.g. through interviews or observations), to tell the story and draw important lessons from that single example.

Example(s): Event organisers conduct a comprehensive (i.e. dealing with all or nearly all aspects of something) security case study of a music festival to understand its challenges, such as crowd control issues and unauthorised entry.

Additional remarks and resources:

- Watch: [For another example of a more in-depth case study, watch this video about the 2014 Sony hack \(11:21\). Take notice of the different elements: what, why, when, how, where, and who.](#)

Sources

Explanation: A source is simply where information comes from (i.e. where you got it from). In research, sources can be articles, books, websites, videos, people, or data that provide evidence for your ideas. Not all sources are equally reliable; they can be primary or secondary, and fall into categories like white or grey literature.

Example(s): A company's server logs and network traffic data are digital sources that can be used to investigate a data breach. During the data breach, they can also be exploited as sources for sensitive customer data.

Additional remarks and resources:

- Watch: [You will learn more about sources in the context of research later on. To get a general idea about them, you can watch this video on "How to Evaluate Sources" \(05:11\).](#)

Citing (a source)

Explanation: Citing is how you give credit to the sources you used. It shows you have done proper research, justifies your arguments, and allows you to acknowledge (i.e. give credit to) the ideas of other researchers who analysed the topic before you. Citing properly is a key part of being a responsible researcher and helps you avoid plagiarism. SSMS uses the APA (American Psychological Association) style, which is commonly used in social sciences. APA style typically involves noting the author's last name and the year of publication directly in the text.

Example(s): There are two main ways to do this:

- Narrative citation: You incorporate the author's name directly into your sentence as part of the narrative. The publication year follows in parentheses. Example: Hardiman and Wong (2022) reported that on July 1st, 2002, two aircraft crashed above Überlingen, Germany, causing 71 deaths.
- Parenthetical citation (also known as in-text citation): You include the author's last name and the publication year in parentheses at the end of the sentence or clause where the information is used. Example: A cost-benefit analysis identifies and evaluates the net benefits associated with different ways of achieving goals compared to their costs (Sassone & Schaffer, 1978).

Additional remarks and resources:

- ☐ Read: [Citing sources in APA style here.](#)
- ☐ You will learn more about sources and how to cite them in ART.

Primary sources

Explanation: Raw data and first-hand evidence that has not been interpreted by someone else. This includes interview transcripts, documented observations, raw statistical data, equipment logs, some (research-based) academic/scholarly articles, and some (research-based) government reports.

Example(s): For an accident investigation, primary sources would include security camera footage, equipment logs, and direct interviews with eyewitnesses.

A bank has experienced a data breach, and the cybersecurity team is tasked with investigating the breach's impact. In this scenario, primary sources include first-hand evidence gathered through transcripts of interviews with IT staff who were involved in discovering and responding to the breach, as well as any individuals who may have detected unusual activity on their accounts. Data on login attempts and access are also crucial primary sources for the investigation.

Additional remarks and resources:

- ☐ Watch: [Examples of the differences between primary and secondary sources \(08:42\).](#)

Secondary sources

Explanation: Second-hand information and interpretations of primary sources. This includes textbooks, articles that explain, review, analyse, or otherwise add value to previous research, histories, and biographies.

Example(s): For research on a historical espionage case, secondary sources would include textbooks and articles written by historians who have already analysed the original intelligence reports and witness testimonies (the primary sources).

A government security analyst is tasked with evaluating the effectiveness of a particular national security policy aimed at countering cyberthreats. Secondary sources include academic articles, policy briefs, and expert analyses written by scholars and security experts. Primary sources include government reports, legislative documents, and cybersecurity incident data.

Additional remarks and resources:

- ☐ Watch: [Learn more about these different types of sources in "The Differences Explained" \(04:12\).](#)

White literature

Explanation: An umbrella term for traditionally published, peer-reviewed academic works. This includes scholarly articles in academic journals and books published by academic presses (not all books are considered white literature).

Example(s): Peer-reviewed journal that publishes white literature: *Global Environmental Politics* examines relationships between global political forces and environmental change, such as the implications of local-global interactions. Each issue of the journal contains several full-length research articles on a broad range of issues, from water to waste management to climate change.

Additional remarks and resources:

- ☐ Read: [Read more about traditionally published sources, or "white literature", here.](#)

Academic articles

Explanation: Formal texts, also known as scholarly articles, studies, or papers, written by experts and published in academic journals after a rigorous peer-review process. They share new knowledge and are typically structured with an abstract, introduction, methodology (including the applied research design), results, and conclusion, and include citations to previous research. They also include citations and references to other academic works that have been written before.

Example(s): The excerpts used in preparation for the CT2 are all part of academic articles.

Additional remarks and resources:

- ☐ Read: [To read this in other words, click here.](#)

Peer-review

Explanation: The formal process where experts in a field (as close as possible to the topic of the work, i.e. peers) evaluate a research paper before it can be published in an academic journal. The author receives feedback and must often make corrections, sometimes through several rounds. This process is a standard for quality control in scholarly publishing and can take months or even a year to complete.

Example(s): A study on workplace safety is submitted to a sociology journal. Experts in sociology and workplace dynamics will peer-review it to assess its methodology and findings before it is published. This rigorous evaluation helps ensure the quality and validity of the research in addressing safety issues within a social context.

The Federal Aviation Administration (FAA) often engages in peer-review when conducting research on aviation safety and air traffic control systems. Experts from academia, industry, and government agencies assess the research to enhance aviation safety practices.

Additional remarks and resources:

- ☐ Watch: [Still feeling a bit confused about this? Watch "Peer Review in 3 minutes" \(03:15\).](#)

Grey literature

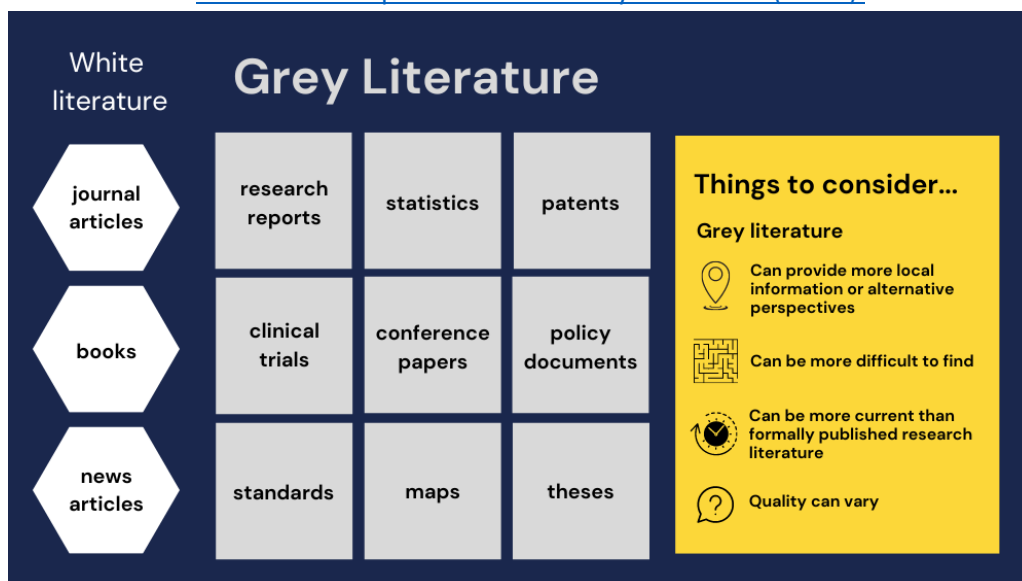
Explanation: An umbrella term that includes materials and research produced by organisations outside of traditional academic publishing. This includes technical reports from government agencies, theses, conference proceedings, and corporate reports. While this information can be very

valuable, it does not go through the same peer-review process as academic articles. They may also have fewer citations and references to previous research or academic works. Abstracts may or may not be included, and are often replaced with an executive summary. The structure of these materials can vary widely depending on the source, since they are not as standardised as academic articles. The term grey literature was coined to distinguish this type of literature from the so-called "white" literature: the traditionally published academic literature that was historically printed on white paper.

Example(s): A report from the World Health Organization (WHO) about a global health crisis, a SSMS thesis, or a technical report from a government agency. These materials may not be published in academic journals but still provide valuable information and insights.

Additional remarks and resources:

- Watch: [See more examples in "What is Grey Literature" \(02:57\).](#)



Part 3: Planning your investigation

Applied research design

Explanation: An applied research design (ARD) is the strategic blueprint for a research project that aims to solve a specific, practical problem. Think of it as the detailed route you plan in your GPS before starting a journey; it ensures you know where you're going and the best way to get there. An effective ARD consists of four core components that must be perfectly aligned:

- **Research problem:** Your starting point. This is the specific, real-world issue or knowledge gap that your research will address.
- **Research objective:** Your destination. This defines the concrete goal, deliverable, or outcome of your project.
- **Research question:** The main route you will take. This is the single, overarching question that your entire project is designed to answer in order to meet your objective.
- **Sub-questions:** The step-by-step directions. These are smaller, focused questions that break down the main research question into manageable parts. Answering all the sub-questions will collectively answer your main research questions.

All four parts must be logically connected, forming a clear and consistent link throughout your research; this is often called the "golden thread" of your research. The questions must directly address the objective, and the objective must be a direct response to the problem. Such a solid ARD will help you present defend your research when faced with supervisors, clients, or other relevant parties.

Example(s): See figure 6 for visual example of the four aligned components of an ARD.

Additional remarks and resources:

- See lecture 3 and workshop 3 slides for examples of what is expected of you at this stage in SSMS.

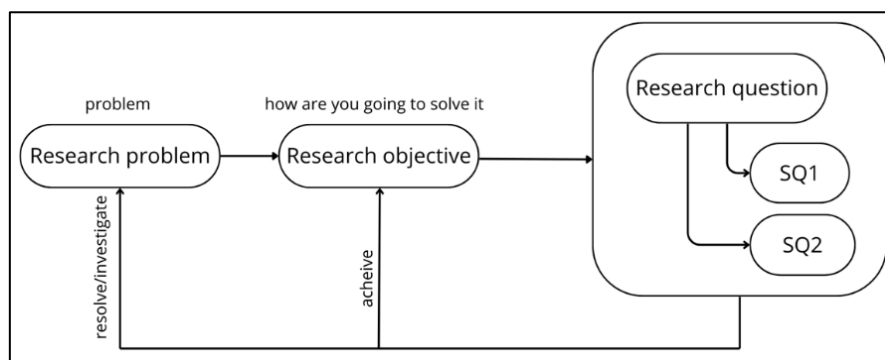


Figure 6. Applied research design

Demarcation

Explanation: The process of setting clear boundaries for your research to make it focused and manageable. A well-demarcated project is crucial because it clarifies your priorities from the start and keeps your investigation on track. The process happens within your ARD (see previous entry), where you narrow a broad topic down into a specific project.

Review the checklist to ensure **every statement and question in your ARD:**

- ☒ **Is focused** and contains **only one** specific safety/security phenomenon? (i.e. the one issue you need to solve, it should be the same one throughout the entire ARD)
- ☒ **Is relevant** and **connects to society and safety/security practice**? Includes the specific context of the case study? (i.e. include as many 6W problem analysis framework answers as relevant)
- ☒ **Follows an applied research perspective**? Does it focus on practical applications to a real-world problem? Does it state what you want to do (e.g. assess, decrease, compare, etc.)?
- ☒ **Is clear and unambiguous in language**? (and does the terminology align with the 6W problem analysis answers?)
- ☒ **Is it unbiased**? Does it use neutral language and avoid leading the reader?
- ☒ **Is it ethical**? Does it consider the safety and dignity of everyone involved? (see "Research ethics")
- ☒ **Is it feasible**? Can it be realistically completed with the available time and sources?
- ☒ **Is it complex enough**? Does it require actual research and analysis (more than a simple search)?
- ☒ **Is it open-ended**? (For questions) If it can be answered with yes/no, it is not open-ended.
- ☒ **Is it single-topic**? (For questions) Does it avoid asking two things at once (i.e. not "double-barrelled")?

Example(s): "How can crowd crushes [SSMS phenomenon], such as the 2022 Halloween disaster in Seoul [case study details], be prevented [applied perspective] during future South Korean celebrations [case study details]?"

Additional remarks and resources:

- ☐ See lecture 3 and workshop 3 slides for examples of what is expected of you at this stage in SSMS.

Types of research questions

Explanation: In applied sciences, the type of research question you choose is crucial because it defines the practical goal of your project. Your question determines whether you are trying to understand a problem, measure its effects, propose a solution, or predict a future outcome. It is the first step in creating a research design that can deliver a useful, real-world result for a client or organisation.

While all types of questions can be used, applied sciences most commonly rely on descriptive, evaluative, and prescriptive questions because these are directly focused on assessing current situations and recommending practical improvements.

Example(s):

Exploratory question	
Purpose: To investigate a new or understudied topic where little is known. It's used to generate initial insights.	Example: How do Dutch university students perceive public safety in major train stations in the post-COVID-19 era?
Descriptive question	
Purpose: To provide a detailed account of a known situation or phenomenon. It usually begins with "what" or "how" and seeks to understand the nature, characteristics, or functions of a specific subject.	Example: What are the current safety protocols in place for handling hazardous chemicals at the Philips High Tech Campus in Eindhoven?
Defining question	
Purpose: To establish a clear and precise meaning for a key concept you are using in your research, to ensure a common understanding.	Example: How can the concept of "cybersecurity resilience" be defined for Dutch small and medium-sized enterprises (SMEs) in the context of increased remote working?
Interpretive/Explanatory question	
Purpose: To understand the "why" behind a phenomenon, seeking to understand its meaning, causes, or the reasons for its existence. Such questions can also focus on understanding how individuals or groups make sense of their experiences, actions, or social contexts.	Example: Why did the success rate of phishing attacks in the Finance department of the Dutch Ministry of Justice differ significantly from the rate in the Human Resources department during the last fiscal year?
Comparative question	
Purpose: To analyse the similarities and differences between two or more cases, groups, or situations. These questions often	Example: How do the incident reporting procedures for security personnel at

include the words "compare," "contrast," or "differences."	Schiphol Airport compare to those at the Port of Rotterdam?
Evaluative question	
Purpose: To assess the effectiveness, impact, value, or success of something (like a policy or implementation) against a set of criteria. It aims to determine, for example, the extent to which the intended outcome was achieved.	Example: To what extent was the new neighbourhood watch app effective in reducing reported bicycle thefts in the Schilderswijk neighbourhood of The Hague during its first year of implementation (2024)?
Prescriptive question	
Purpose: To propose solutions or recommend future actions for a specific issue.	Example: What communication strategies should THUAS implement to increase cybersecurity awareness among its first-year students?
Predictive question	
Purpose: To try to forecast future outcomes based on current trends. It often uses statistical data analysis.	Example: What is the likely impact of widespread 5G technology adoption on the effectiveness of real-time crowd monitoring systems at major public events in Amsterdam in the next 5 years?

Additional remarks and resources:

- ☐ [Video about exploratory research \(04:08\).](#)
- ☐ Read: Verhoeven, 2019, p.78 (online 3.2.1).

Research limitations

Explanation: Every research project has limitations—constraints or challenges that might affect the results. They are an accepted part of the research process, and should be disclosed in full transparency. Because of this, it's crucial to identify these limitations, explain their potential impact on your specific project, and (if possible) describe how you plan to avoid them, minimise their impact, or address them. If a limitation cannot be minimised or addressed, you must still be transparent about it. There are a lot of potential research limitations. The most common occur when: (1) data availability is scarce or insufficient, (2) necessary data access is restricted, (3) the sample size is small, (4) sample population is not representative of the larger population, (5) the time constraints of the project are such that there is not enough time to conduct the research thoroughly, or (6) budget constraints are present, considering potential costs of the project. However, if you take a look at the "Research ethics" entry, you will see that there are many other potential limitations related to research Honesty, Independence, Transparency, Responsibility, and Scrupulousness.

Example(s): A study on the safety risks of a brand-new type of drone might be limited by the lack of available accident data (i.e. data availability). To mitigate (i.e. minimise the impact of) this, the researcher could supplement the study with data from similar drone types or use simulation models. This limitation often occurs when a problem is still relatively recent, or happens very rarely.

A 5-month study evaluating a 5-month safety programme might be limited by time constraints, as the full impact might only be visible after the programme has ended. The researcher could address this by focusing on short-term indicators or comparing the programme to similar, completed programmes elsewhere.

A researcher has the responsibility to protect participants from physical, psychological, or social harm (see "Research ethics" entry). To uphold this duty of care, a study investigating safety protocols in extreme work environments excludes high-risk demographic groups (e.g. pregnant workers, individuals with pre-existing conditions) to prevent potential harm. However, this creates a sample population bias, because the findings cannot be generalised to excluded populations, which is a limitation of the study.

Additional remarks and resources:

- ☐ See the entry for "Research ethics" to read more about research Honesty, Independence, Transparency, Responsibility, and Scrupulousness. What other potential research limitations can you come up with, based on that information?
- ☐ See lecture 3 and workshop 3 slides for more information and examples of research limitations, and how to briefly justify them.

Generalisability

Explanation: This refers to how well the findings from your specific study can be applied to a broader context, beyond the specific conditions of the research (e.g. after studying a certain group of people, can the results be applied to the larger population? Can the results be applied to another country, or similar situations? To what extent?). Ideally, research aims for high generalisability so the results are

widely useful. This means that you, as a researcher, must aim to make your research findings as relevant to other groups of people or situations as possible. However, not all research findings are equally generalisable, this depends heavily on the research design, especially the representativeness of the sample population. Carefully designing your study increases confidence in its generalisability. When you are reading or evaluating the quality of a particular (existing or future) research project, pay attention to its potential generalisability. It is a key factor in determining the usefulness and impact of those research findings. It is often connected to the concepts of representative and unrepresentative sample population.

Example(s): In a study on airport security, a high level of generalisability would mean that the results from improving security checks at one airport could be applied to other airports with similar conditions. If the research is conducted using a representative sample, then the solutions tested for one location can be expected to work in other, similar environments, within reason (e.g. security checks are effective at Schiphol Airport, generalisability asks if these findings would also apply to Rotterdam The Hague Airport or Eindhoven Airport, considering their similarities and differences). This makes the research results more widely applicable to different situations and, thus, preferable in terms of efficiency.

A retail chain conducts a security audit at one of its store locations to assess the effectiveness of its security measures, such as surveillance cameras and alarm systems, in reducing theft incidents. The challenge is to determine whether the security measures found effective in this specific store can be generalised to other stores within the chain. To evaluate generalisability, the retail chain may conduct similar security audits at multiple store locations across different regions. If the effectiveness of the security measures holds true across various store settings (i.e. a representative sample and a large sample size is used), the company can reasonably conclude that these security practices can be applied more broadly to enhance security throughout the entire retail chain (i.e. the larger target population).

Additional remarks and resources:

- See the entries for "Population vs sample population" and "Representative vs unrepresentative sample" for more information about generalisability and the factors that can contribute to it.

Participant vs respondent

Explanation: In general, both terms refer to individuals providing data for research, but they have slightly different nuances. While the terms are sometimes used interchangeably, being precise helps clarify your research method.

- Respondent is typically used for someone who answers specific, often closed-ended questions, like in a survey. Their input is often used for quantitative analysis.
- Participant is generally used for someone who actively takes part in the research process, such as in interviews, focus groups, experiments, or ethnographic observation. Their input is often used for qualitative analysis.

Example(s): In a research project focused on online security, the respondents are users of an online banking platform. Quantitative data, including a questionnaire about their login behaviour, password practices, and responses to security prompts, is studied to evaluate the effectiveness of the platform's

security measures. Some of the respondents are invited to become participants in a qualitative follow up, in the form of an interview to better understand their impressions of online security. The findings are used to enhance the platform's security features and better protect users' financial information.

Additional remarks and resources:

- See the entry for "Research ethics" to read more about how you need to take participants and respondents into consideration when developing your applied research design.

Population vs sample population

Explanation: In the context of research, population, also known as the target population, represents the general group that is being researched about in the study. That is, the larger group that researchers want to draw conclusions about. Once that population has been clearly defined, researchers attempt to select a smaller, more manageable "sample population" that is considered representative of this population of people to whom results will be generalised. This is done so that the researchers can focus on collecting data from only a portion of the larger population. In studies that use statistics to analyse results, or which are designed to be generalisable, a large sample size is critical, as generally the larger the number in the sample, the higher the likelihood of a representative distribution of the population. However, often researchers have to work with a smaller sample population size (e.g. only a few people are available to you, and/or have enough expert knowledge on the issue, and/or the research methods used, such as interviews, are very time consuming). To increase your chances of getting the data you need and making generalisable claims, you should aim for a sample population that is as representative as possible of the larger population. Sometimes, practical limits mean working with smaller or less representative samples, which then becomes a research limitation.

Note on vulnerable populations: When your research involves groups considered vulnerable (e.g. children, refugees, victims of crime, individuals with cognitive impairments, marginalised populations, etc.), extra ethical care is required. This includes ensuring truly informed consent (be as transparent as possible, including about any risks they might take by taking part in your research), protecting anonymity rigorously (anonymise all data, safeguard any sensitive data that contains identifiable information behind 2FA and encryption), and minimising any potential harm or burden on them (including potential political or social discrimination, time spent on the research, emotional harm, financial costs, etc.). Your research design and ethical review must explicitly address how you will safeguard their wellbeing and how your research benefits these participants/respondents.

Example(s): In a study focused on fire evacuation preparedness in schools in The Hague, the population has been defined as "students and staff at all primary education schools in The Hague". The sample population consists of students and staff in one primary school in the Hague. The researchers aim to understand the level of preparedness, knowledge of evacuation procedures, and the presence of safety equipment among this sample population. By studying this (hopefully representative) sample of primary school students and staff, the findings can be generalised to improve fire safety practices in similar primary school settings in The Hague.

In a study evaluating the effectiveness of workplace safety training programmes in a large manufacturing company franchise with four locations, researchers aim to select a sample of employees. They decide to focus their research on one location of the franchise. That is the sample

population they will study, so that they can generalise findings to the other manufacturing company locations. Since resources and protocols should be similar at the different locations, this sample population seems to be adequate.

Additional remarks and resources:

- See the entries for "Generalisability" and "Representative vs unrepresentative sample" for more information about how your choice of sample population can affect the usefulness of your research findings.
- Watch: [If you are still struggling with these concepts, you can watch this video on "Defining your Population and Sampling Strategy" \(05:49\).](#)

Representative vs unrepresentative sample

Explanation: A representative sample population is one whose key variables and characteristics closely match the characteristics of the larger target population you are studying (e.g. similar proportions in age, gender, income, etc.). Findings from a representative sample can usually be generalised to the larger target population. An unrepresentative sample, on the other hand, is a portion of the population that does not accurately reflect the characteristics of the overall population. Because of this, the findings you draw from this sample cannot be generalised to the population you selected. When a sample is not representative, it can demonstrate sample bias, such as overrepresentation or underrepresentation of certain groups. Overrepresentation in an unrepresentative sample occurs when one or more groups are disproportionately represented compared to their actual presence in the population. In contrast, underrepresentation would occur when one or more groups are not represented or are insufficiently represented compared to their actual presence in the population. You can still try to take some limited insights from this data, but you must be transparent about the lack of representativeness in your sample.

Example(s): A study on cybersecurity awareness among users of the online platform "Talkers" initially called for participants to sign up for the research through a post on the platform. This creates an unrepresentative sample because very active users are overrepresented, while less active users are underrepresented. The findings would likely overestimate awareness, because more active users might be more aware of cybersecurity measures. To fix this, researchers reached out through email as well and actively selected participants from various user segments (e.g. activity level, age, location) to create a representative sample.

A study wants to examine the public opinion on neighbourhood policing practices in a specific neighbourhood in The Hague, including citizens' sense of security within their community, and their interactions with local law enforcement. They decide to focus on a smaller group of people who live in that specific neighbourhood, and generalise their research findings to the whole neighbourhood. For the sample population to be representative, it must include participants with diverse backgrounds, considering key socioeconomic variables such as income, education level, employment, but also age, religion, and ethnic background. By doing so, the research results (based on a smaller group of people) can be generalised to the entire population (the whole neighbourhood). This allows policymakers to make informed decisions about new community policing strategies in a way that match the needs of the entire population of the neighbourhood.

Additional remarks and resources:

- Watch: [Learn more about sample selection bias. Without seeing the solution, can you figure out where more armour should be added? "Do You Really See the Whole Picture?"\(up until 04:20\).](#)

Independent vs dependent variables

Explanation: In research, a variable refers to a characteristic or aspect that can be measured, manipulated, or counted in a research study. They are used to explore relationships between different factors. Examples of variables include age, political beliefs, gender, education level, income, and exposure (or no exposure) to a particular intervention/measure. Essentially, any characteristic or trait that can vary from one person/situation to another, or for one person/situation over time, can be considered a variable. There are different types of variables to keep in mind.

The independent variable is the one you are able to change during your research (which is why it is also known as the “manipulated”, “input”, or “cause” variable). This is the variable you manipulate or change to see what happens.

The dependent variable is the one that you are interested in measuring (it is also known as the “responding”, “output”, or the “effect” variable), so that you can collect useful data for your study. This DV changes due to, at least in part, the impact of the IV(s). In other words, the value of the DV changes depending on the value of the IV(s).

Example(s): In a safety study conducted in a manufacturing plant, the dependent variable is the “frequency of safety incidents every month”. This variable changes based on the impact of various factors, including “safety training provided” (e.g. 0 hours vs 10 hours). Researchers want to determine if *increased* “safety training” (which now becomes our IV) results in a *decrease* in “frequency of safety incidents every month” (DV).

Additional remarks and resources:

- [Watch "Independent and Dependent Variables" for simpler examples of different types of variables \(08:36\).](#)
- For more explanations on this, see sheets available on BS (van Tulder, 2018, pp.80-81; Verhoeven, 2019, pp.238-239) and via the THUAS e-Library (Verhoeven, 2011, section 8.1.1).

Control variables

Explanation: There is also the control variable, also known as the “constant” variable. When testing if A causes B, you must ensure that nothing else is interfering. The control variable(s) must remain unchanged throughout your study for accurate results, so you can be sure that the dependent variable is not being affected by the control variable(s). If you do not intentionally control these, you cannot be sure your results are valid.

Example(s): (Continuing example from previous concept) The control variable(s) in a study like this is another variable that can also affect the frequency of safety accidents, such as the use of personal protective equipment. By making sure that *all* participants of the study have access to the *same* PPE,

researchers will be able to more accurately account for the changes caused *only* by the *increase* of "safety training provided" (independent variable). If one group has better helmets than the other, you will not know if the lower accident rate is due to your training (independent variable) or the difference in helmets. You must ensure everyone has the same PPE (control this variable) so the only difference is the training.

Additional remarks and resources:

- [Watch "Research Variables 101: Dependent, Independent, Control Variables & More \(With Examples\) \(11:36\)."](#)

Confounding variables

Explanation: A confounding variable is an external, often hidden factor that can affect both the independent and dependent variables (i.e. affecting the cause-and-effect relationship), without you being aware of this. This can lead to misleading conclusions in research, as well as errors, as it makes it difficult to know whether the effects you are seeing are truly due to the independent variable or the confounding variable.

Example(s): (Continued from previous concept) A confounding variable could be "workplace culture". If employees' attitudes towards safety procedures are already improving due to an unrelated variable, such as management style, this might also impact "frequency of safety incidents every month" without you noticing. Imagine a new manager is hired who is very strict. This new manager might make people take the training (independent variable) AND force them to be safer (dependent variable), regardless of the training. You might think the training worked really well, but actually, it was mostly/just the new manager. This effect is not being measured, and it makes it difficult to attribute the effect of the study solely to the increase "safety training provided" intervention.

Additional remarks and resources:

- [Watch "Research Variables 101: Dependent, Independent, Control Variables & More \(With Examples\) \(11:36\)."](#)

Mediating variables

Explanation: You can also spot mediating (aka intermediary or intervening) variables, which reveal that the connection between the supposed dependent and independent variable is not direct, but occurs via another variable in between. It is a stepping stone between the dependent and independent variable, helping to explain how or why the relationship happens.

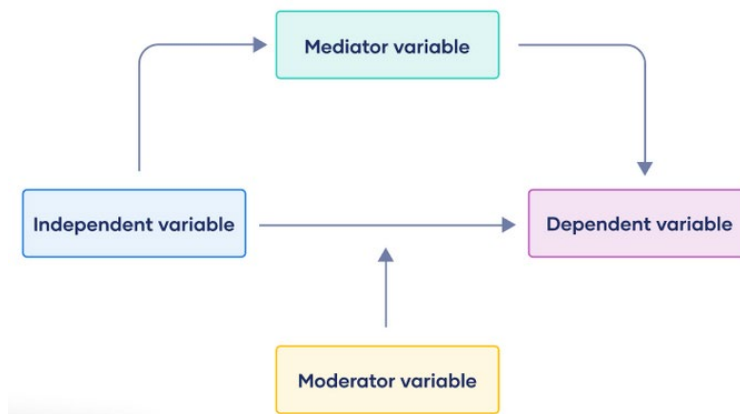
Example(s): (Continued from previous concept) In this same study, an mediating/intermediary variable could be "changes in safety practices", where the training leads to better safety practices first, which in turn can reduce incidents. In other words, "safety training provided" (IV) leads to "better safety practices" (mediator), which in turn leads to a decrease in the "frequency of safety incidents every month" (DV). The training works through the practices.

Additional remarks and resources:

- [Read more about moderating and mediating variables here.](#)

- For more explanations on this, see sheets available on BS (van Tulder, 2018, pp.80-81; Verhoeven, 2019, pp.238-239) and via the THUAS e-Library (Verhoeven, 2011, section 8.1.1).

Mediator and moderator variables



Moderating variables

Explanation: You will also encounter moderating (aka interfering) variables, which affect the strength or direction of the relationship between your dependent and independent variable (i.e. they have some influence between the cause and effect by, for instance, making the effect more or less substantial). It explains for whom or under what conditions the relationship holds. In other words, why the cause-effect relationship is strong for one group and weak for another.

Example(s): (Continued from previous page) In this same study, a moderating/interfering variable could be “employee experience level”, which participants are obtaining at their own different individual pace. This variable clarifies how an increase in “safety training provided” impacts the “frequency of safety incidents every month” differently for each person, team, or department, by explaining one of the steps in the cause-and-effect process that are making the impact on the effect more/less effective. For example, maybe training drastically reduces accidents for newly hired employees, but has almost no effect on veterans (who already know it). This could mean that, after the training is provided, teams with a lot of new hires might see a larger drop in safety incidents than teams with a lot of veterans. The “employee experience level” (moderating variable) makes the relationship between the training provided (IV) and decreased safety incidents (DV) stronger or weaker.

Additional remarks and resources:

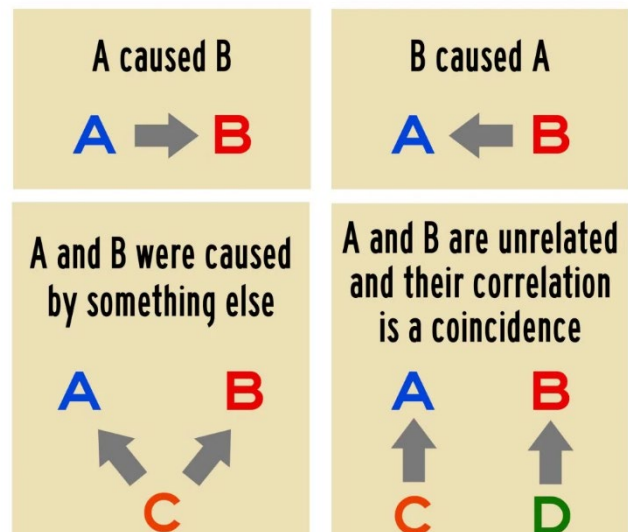
- [Read more about moderating and mediating variables here.](#)
- For more explanations on this, see sheets available on BS (van Tulder, 2018, pp.80-81; Verhoeven, 2019, pp.238-239) and via the THUAS e-Library (Verhoeven, 2011, section 8.1.1).

Relationships between variables

Explanation: Understanding the relationship between variables and events helps us predict what could happen in the future, and also reflect on why things occurred in the past. Correlation is a common statistical data analysis that allows us to better understand relationships between data, as it measures the degree of relationship between variables in a sample. It is an important piece of the puzzle when you are looking for a clear relationship between variables. However, just because two variables move together (correlation) does not mean one caused the other (causation). When an independent variable, "A", is correlated with a dependent variable, "B", there are a few possible reasons:

- (1) A causes B (causal relationship), if there is proof that a change in A, and nothing else, directly creates a change in B. This usually requires a strictly controlled experiment to prove, where other variables are controlled and accounted for.
- (2) B causes A (reverse causality).
- (3) Another variable, such as a confounding variable C, causes both A and B, even though A and B are not directly related.
- (4) There is no relationship at all, as sometimes two completely unrelated things are correlated by coincidence. This is known as a spurious correlation.

Possibilities if A is correlated with B



Example(s):

- A fire department wants to prove that faster response times reduce property damage. They systematically improve response times (IV) while keeping equipment and team size constant (control variables). They observe a consistent 25% reduction in property damage (DV). Because they controlled for other factors, they can claim a causal relationship: "Reducing response time causes less damage."
- A researcher concludes that "firefighters cause damage," because every time a large group of them shows up, the building suffers more damage than when fewer of them show up. The reality is that the severity of the fire (which leads to high damage, variable B) causes the dispatcher to send more firefighters (A). The damage is not caused by the firefighters; the massive response is caused by the massive fire. The arrow of causality points backward.
- We observe a correlation between "ice cream sales at the station" (variable A) and "number of brush fires (variable B). One might think that firefighters eating ice cream leads to fires. The reality is a confounding variable (variable C), "summer heat", is affecting both A and B. A and B are statistically related, but there is no direct link between them.
- We find a spurious correlation between the "fire department's chief's age" (variable A) and the "national price of water" (variable B) over the last 20 years. Both have steadily increased. The chief gets older every year (variable C, time). Inflation might make water more expensive

every year (variable C). They both go up, but the chief's age has absolutely no impact on the national water market. It is a pure coincidence.

Additional remarks and resources:

- [Watch "Correlation vs Causation" \(02:10\) for a simpler example of relationships between variables.](#)
- [Read: More spurious correlation examples.](#)

Hypothesis

Explanation: When you start your research, you need to be able to define your general assumptions. These can change a lot over time, but they provide you with a direction to start your research. These are called hypotheses. A hypothesis is a formal statement or prediction of what the researcher expects to find. It needs to be testable (that is, you need to be able to verify it through research) and it must try to predict a causal relationship between variables (e.g. "if I do X, then Y will happen"). You will notice that your hypothesis is the "predicted answer" that is closely related to your research question.

Example(s): In a manufacturing company, the management has introduced a new safety training programme for its employees (IV), emphasising the importance of using personal protective equipment (PPE) like helmets and gloves, to see if it reduces accidents (DV). Their research question is "to what extent does the implementation of the new safety training programme lead to a reduction in workplace accidents?". Their hypothesis (prediction) is that "the implementation of the new safety training programme, with a focus on PPE usage, is causally linked to the observed reduction in workplace accidents."

An online banking platform wants to increase password complexity (IV) to stop hackers (DV). Their research question is "what is the effect of increased password complexity requirements on the number of unauthorised access attempts?". Their hypothesis (prediction) is that "if the online banking platform has more complex password requirements, then there will be a significant decrease in successful unauthorised access attempts."

Additional remarks and resources:

- [Are hypotheses still eluding you? Watch "6 Steps to Formulate a STRONG Hypothesis" \(03:30\).](#)

Reliability

Explanation: Reliability is about consistency. It refers to the degree to which a study, measure, method, or variable produces the same results when applied to similar scenarios over and over again. Ask yourself: does it measure in a consistent and stable way? If the measuring instrument (e.g. survey or a scale) is reliable, then repeating it with similar groups would offer similar results. If this does not occur, there is no reliability (you cannot depend on it to provide trustworthy results over time). Reliability is a prerequisite for validity. If your tool is broken (unreliable), it cannot possibly give you the correct answer (validity). You will learn more about how this works in practice when you study specific research methods in other courses.

Example(s): Reliability in a non-research context: In a high-rise office building, a fire alarm system is installed to alert occupants in case of a fire emergency. The reliability of the system is crucial for ensuring the safety of building occupants. A fire alarm is reliable if it goes off every single time the button is pressed. If you press it 10 times and it only rings 6 times, it is unreliable. You cannot depend on it during emergencies.

Reliability in a research context: In a large corporation, HR conducts an annual satisfaction survey. If the survey produces consistent results regarding "perceived safety" year after year (assuming the actual safety has not changed), the survey tool is reliable. This allows HR to track real trends rather than random noise.

Additional remarks and resources:

- [Why do we need to know about reliability and validity? Watch "Qualitative research the basics" \(06:57\) for a simple explanation.](#)

Validity

Explanation: Validity is about accuracy. It refers to the degree to which a study, measure, or method accurately reflects the specific concept that the researcher is attempting to measure. Ask yourself: does it measure what it is supposed to be measuring? A method can be reliable (consistently measuring the wrong thing) but not valid.

External validity refers to the extent to which the results of a study are generalisable or transferable beyond the setting of that specific study. Ask yourself: can the study be applied to the "real world" beyond this specific study? Can it be applied in other relevant contexts (i.e. are the study findings of a study conducted in The Hague also applicable to similar cities, like Leiden)?

Internal validity refers to the rigor with which the study was conducted (e.g. the research design is solid, methods were carefully followed, and decisions concerning what was and was not measured were clear, etc.). Ask yourself: was the study conducted correctly? This is also about the extent to which researchers, like yourself, have taken into account alternative explanations for any causal relationships they explore (see entry on variables). For example, you must consider whether the results were due to the manipulation of the independent variable rather than other factors such as intervening variables. You will learn more about how this works in practice when you study specific research methods in other courses.

Example(s):

- External validity: A research study evaluates the effectiveness of an emergency response training programme for first responders in a specific urban area. If the training program's effectiveness can be generalised (i.e. it can be applied beyond the study) to other urban areas, populations, or times, it demonstrates high external validity.
- Internal validity: A cybersecurity team investigates a data breach to determine how it occurred. If they can prove that a specific malware caused the breach, and they successfully rule out human error or insider threats (confounding variables), their conclusion has high internal validity. This means it is measuring what it was supposed to be measuring (i.e. it had

a high accuracy within its own context). This is usually related to having control over variables and avoiding confounding variables.

Additional remarks and resources:

- [Want to see more examples? Watch "What Are Validity & Reliability In Research" \(08:25\).](#)

Conceptual framework and conceptual model

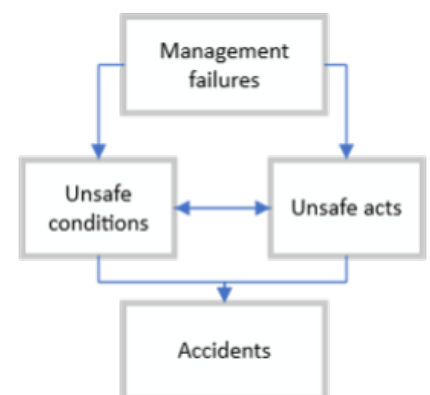
Explanation: Imagine you are about to start a research project. You have a general, vague topic in your head, like "accidents." Before you can measure it, you need to pin down its meaning. This process starts with concepts, which are the abstract ideas you want to study (e.g. like these concept lists).

To make these concepts usable, you must demarcate them (i.e. define and clarify what you mean when you use them). This means drawing a boundary around your concept and stating exactly what it means in the context of your research. You often rely on pre-existing theories to help you define these boundaries (such as the Accident Theory below), since they already wrote down how these concepts can be demarcated, which saves a lot of time and effort. Once defined, you break the concept down into variables (the specific aspects that can change or vary) and indicators (the concrete things you can actually count or measure). Often, theories give you some idea of how to do that as well. These elements are outlined in your conceptual framework. Think of a conceptual framework as the "script" or the theoretical argument. It is the written explanation that links your variables together. It explains why you believe variable A affects variable B.

Then, you have the conceptual model. If the framework is the script, the model is the "map." It is a simplified visual diagram that translates your complex theory into boxes and arrows. The boxes represent your variables, and the arrows show the relationships (e.g. cause and effect). The direction of the arrows matters immensely, as they tell the story of what leads to what (see entry on relationships between variables and hypothesis).

Example(s): Let's say you want to study why accidents happen in a factory. You cannot just measure "accidents" in a vacuum, you need a framework. You choose Accident Theory, which suggests accidents are not random. Accident Theory is based on the idea that accidents are caused by a combination of factors, including unsafe conditions, unsafe acts, and management failures. You break the concept of "accidents" down into three specific variables: unsafe conditions (the environment), unsafe acts (human behaviour), and management failures (the root cause).

Unsafe conditions (i.e. indicators include looking for slippery floor, faulty equipment, lack of safety signs, etc.) create an environment where accidents can occur. Unsafe acts (i.e. indicators include counting how many workers are skipping personal protective equipment, ignoring safety rules, operating machinery without proper training, etc.) by individuals can lead to accidents. Management failures (i.e. failure on the part of management to provide safety training, equipment, supervision),



while not directly involved in the accident causation process, can make the situation worse by creating or maintaining unsafe conditions, or not providing adequate training or supervision that could prevent unsafe acts.

The theory states that management failures do not cause accidents directly. However, poor management leads to a messy factory (unsafe Conditions) and untrained workers (unsafe Acts). Those two things then combine or amplify each other to cause the accident. Can you find the 6 relationships between the variables in the model?³

Additional remarks and resources:

- ☐ [Need a simpler example of a conceptual framework? Watch "How to Develop a Conceptual Framework" \(04:41\).](#)
- ☐ For more explanations on concept demarcation, and many examples on conceptual framework and model building, see sheets available on BS (Verhoeven, 2019, pp.85-92) and via the THUAS e-Library (Verhoeven, 2011, section 3.3-3.4).

³ Tip: Each connection between concepts counts as a different relationship.